

Adnan Harun Dogan

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EDUCATION

 Master of Science in Computer Engineering <i>Middle East Technical University (METU)</i>	2022 – 10 Jan 2025 <i>Ankara, Turkey</i>
 Bachelor of Science in Computer Engineering <i>Middle East Technical University (METU)</i>	2018 – 17 Jul 2022 <i>Ankara, Turkey</i>

EXPERIENCE

 Research Intern <i>Sensing, Interaction, & Perception Laboratory (SIPLab), ETH Zürich</i> - Developed EEG-based cybersickness detection using multitaper spectrum estimation in VR - Achieved 30% improvement over baselines with novel deep learning architectures - Published at IEEE/ICRA 2026	Dec 2024 – Mar 2025 <i>Zürich, Switzerland</i>
 Graduate Research Assistant <i>Image Processing and Pattern Recognition Laboratory (ImageLab), METU</i> - Designed bucketed ranking loss with Sinkhorn-Knopp algorithm for DETR object detection architecture - Improved end-to-end object detection via optimal transport-based loss formulations	Feb 2022 – Present <i>Ankara, Turkey</i>
 Undergraduate Research Intern <i>Cyber-Physical Systems Laboratory, University of California, Irvine</i> - Built multi-modal deep learning system for cardiac and sleep abnormality detection - Designed beat-by-beat arrhythmia classification integrating time-series and spectral features - Published at Computing in Cardiology (CinC) 2021	Jun 2021 – Sep 2021 <i>Irvine, California, USA</i>

PUBLICATIONS

Dogan, A. H. , Demirel, B. U., Holz, C. (2026). Frequency-Weighted Neural Kalman Filters. <i>ICRA 2026</i> . [GitHub]
Demirel, B. U., Dogan, A. H. , Rossie, J., Möbus M., Holz, C. (2025). Beyond subjectivity: Continuous cybersickness detection using eeg-based multitaper spectrum estimation. <i>IEEE/TVCG 2025</i> . [DOI — GitHub]
Yavuz, F., Cam B.C., Dogan, A. H. , Oksuz, K., Kalkan, S., Akbas, E. (2024). Bucketed Ranking Loss for Efficient Ranking-based Training of Object Detectors. <i>ECCV 2024</i> . [ECVA — DOI — GitHub]
Amin M. A., Dogan, A. H. , Kuru E. S., Sever Y., Angin P. (2024). Misuse Detection and Response for Orchestrated Microservices Based Software. <i>AINA 2024</i> . [KAGGLE — PAPER]
Karagoz, P., Cekinel, R. F., Dogan, A. H. , Oktay, B., Ozturk, A. U., Tonay, S. T., Tunel, B. M. (2024). Enhancing Underground Built Heritage Analysis with Text Mining: A Case Study on Cappadocia. <i>Book Chapter 2024</i> . [PAPER]
Sever Y., Dogan, A. H. (2023). A Kubernetes dataset for misuse detection. <i>ITU Journal 2023</i> . [DOI — Paper — GitHub]
Sever, Y., Ekinici, G., Dogan, A. H. , Alparslan, B., Gurbuz, A. S., Jabrayilov, V., Angin, P. (Sep 2022). An Empirical Analysis of IDS Approaches in Container Security. <i>IEEE/SRMC 2022</i> . [DOI]
Demirel, B. U., Dogan, A. H. , Al-Faruque, M. A. (2021). Two Might Do: A Beat-by-Beat Classification of Cardiac Abnormalities using Deep Learning and Domain-Specific Features. <i>CinC 2021</i> . [GitHub — PAPER — DOI]
Buyukbas, E. B., Dogan, A. H. , Ozturk, A. U., Karagoz, P. (2021). Explainability in Irony Detection. <i>DaWaK 2021</i> . [PAPER]
Dogan, A. H. , Dogan, A. (Jun 2021). An Assembled Deep Learning Approach for Flow Field Prediction. [PAPER]

HONORS & AWARDS

 TUBITAK 2247 C STAR Scholarship <i>The Scientific and Technological Research Institution of Turkey (TÜBİTAK)</i>	2022-2023
 Best Paper Award <i>IEEE/SRMC'22</i>	September 2022
 TUBITAK 2247 C STAR Scholarship <i>The Scientific and Technological Research Institution of Turkey (TÜBİTAK)</i>	2021-2022
 TUBITAK 2247 C STAR Scholarship <i>The Scientific and Technological Research Institution of Turkey (TÜBİTAK)</i>	2020-2021

TECHNICAL SKILLS

Skills: Python/PyTorch, Variational and Generative Modeling, Computer Vision, Statistical Machine Learning, Combinatorial Optimization, Reinforcement Learning